

Chapitre 8 : Les modèles linéaires généralisés

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1 Avant de commencer

Ce chapitre reprends les codes du chapitre 8 sur *les modèles linéaires généralisés* du livre **Langage R et Statistiques : Initiation à l'analyse de données** publié par les éditions ENI (2^{ème} édition).

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.0    ✓ readr      2.1.6
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2     4.0.2    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr       1.2.1
— Conflicts — tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
  conflicts to become errors
```

```
library(multcomp)
```

```
Le chargement a nécessité le package : mvtnorm
Le chargement a nécessité le package : survival
Le chargement a nécessité le package : TH.data
Le chargement a nécessité le package : MASS
```

```
Attachement du package : 'MASS'
```

```
L'objet suivant est masqué depuis 'package:dplyr':
```

```
select
```

```
Attachement du package : 'TH.data'
```

```
L'objet suivant est masqué depuis 'package:MASS':
```

```
geyser
```

2 Modèles linéaires classique

2.1 ANOVA et lm

```
lm(Petal.Length ~ Sepal.Length, data = iris) |> anova()
```

Analysis of Variance Table

Response: Petal.Length

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Sepal.Length	1	352.87	352.87	468.55	< 2.2e-16 ***
Residuals	148	111.46	0.75		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
aov(Petal.Length ~ Sepal.Length, data = iris) |> anova()
```

Analysis of Variance Table

Response: Petal.Length

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Sepal.Length	1	352.87	352.87	468.55	< 2.2e-16 ***
Residuals	148	111.46	0.75		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
lm(Petal.Length ~ Species, data = iris) |> anova()
```

Analysis of Variance Table

Response: Petal.Length

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Species	2	437.10	218.551	1180.2	< 2.2e-16 ***
Residuals	147	27.22	0.185		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
aov(Petal.Length ~ Species, data = iris) |> anova()
```

Analysis of Variance Table

Response: Petal.Length

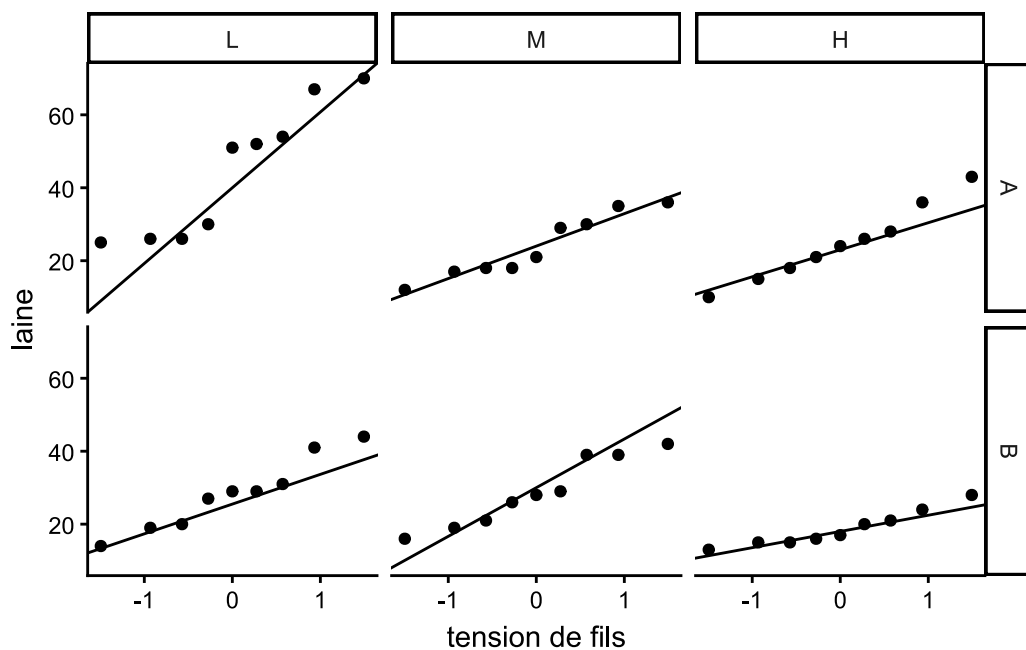
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Species	2	437.10	218.551	1180.2	< 2.2e-16 ***
Residuals	147	27.22	0.185		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

2.2 Test post-hoc de Tukey

vérification de la normalité et de l'homogénéité des données

```
ggplot(warpbreaks) +  
  aes(sample = breaks) +  
  geom_qq() +  
  geom_qq_line() +  
  facet_grid(rows = vars(wool), cols = vars(tension)) +  
  labs(x = "tension de fils", y = "laine") +  
  theme_classic()
```



construction du modèle

```
fm1 <- aov(breaks ~ wool + tension, data = warpbreaks)
summary(fm1)
```

```

      Df Sum Sq Mean Sq F value    Pr(>F)
wool    1    451   450.7    3.339 0.07361 .
tension  2   2034  1017.1    7.537 0.00138 **
Residuals 50   6748   135.0
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

vérification des résidus

```
shapiro.test(fm1$residuals)
```

```

Shapiro-Wilk normality test

data:  fm1$residuals
W = 0.96776, p-value = 0.1535
```

test post-hoc {stats}

```
TukeyHSD(fm1, "tension")
```

```

Tukey multiple comparisons of means
 95% family-wise confidence level

Fit: aov(formula = breaks ~ wool + tension, data = warpbreaks)

$tension
      diff      lwr      upr    p adj
M-L -10.000000 -19.35342 -0.6465793 0.0336262
H-L -14.722222 -24.07564 -5.3688015 0.0011218
H-M  -4.722222 -14.07564  4.6311985 0.4474210
```

test post-hoc multcomp

```
library(multcomp)
summary(
  glht(
    fm1,
```

```

linfct = mcp(
  tension = "Tukey"
)
)
)

```

Simultaneous Tests for General Linear Hypotheses

Multiple Comparisons of Means: Tukey Contrasts

Fit: aov(formula = breaks ~ wool + tension, data = warpbreaks)

Linear Hypotheses:

	Estimate	Std. Error	t value	Pr(> t)	
M - L == 0	-10.000	3.872	-2.582	0.03359	*
H - L == 0	-14.722	3.872	-3.802	0.00113	**
H - M == 0	-4.722	3.872	-1.219	0.44737	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)

test post-hoc {rstatix}

```
rstatix::tukey_hsd(fm1)
```

```

# A tibble: 4 × 9
  term      group1 group2 null.value estimate conf.low conf.high  p.adj
* <chr>   <chr>   <chr>      <dbl>    <dbl>    <dbl>    <dbl>  <dbl>
1 wool     A       B           0    -5.78   -12.1     0.573 0.0736
2 tension L       M           0   -10.00  -19.4    -0.647 0.0336
3 tension L       H           0   -14.7   -24.1    -5.37 0.00112
4 tension M       H           0    -4.72  -14.1     4.63 0.447
# i 1 more variable: p.adj.signif <chr>

```

2.2.a présentation des données en assignant

assignation des lettres en fonction du test post-hoc de Tukey

```

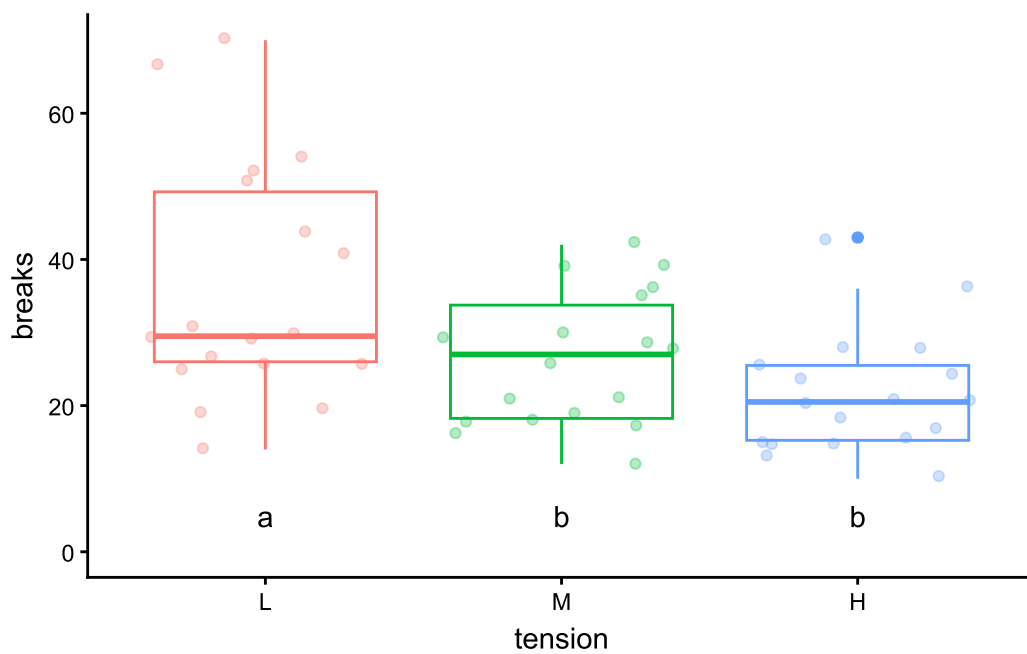
lettre_significative <- tibble(
  lettre = (

```

```
summary(
  glht(
    fm1,
    linfct = mcp(
      tension = "Tukey"
    )
  )
) |>
  cld())$mcletters$Letters,
  tension = names(lettre)
)
```

réalisation du graphique

```
ggplot(warppbreaks) +
  aes(x = tension, y = breaks, color = tension) +
  geom_boxplot() +
  geom_jitter(alpha = 0.3) +
  geom_text(
    aes(x = tension, label = lettre),
    data = lettre_significative,
    color = "black",
    y = 5
  ) +
  theme_classic() +
  coord_cartesian(ylim = c(0,70)) +
  theme(legend.position = "none")
```



```
ggsave("img/08Rl05N.png", width = 14, height = 10, units = "cm")
```

3 glm

3.1 régression logistique

```
library(survival)

donnees_survie <- jasa |>
  mutate(
    survie = fustat |> as.factor(),
    pontage = surgery |> as.factor()
  )
```

modèle

```
m1 <- glm(
  formula = survie ~ futime + pontage + age,
  data = donnees_survie,
  family = binomial
)
car::Anova(m1, type = "2")
```


Analysis of Deviance Table (Type II tests)

Response: survie

	LR	Chisq	Df	Pr(>Chisq)
futime	23.9238	1	1.002e-06	***
pontage	0.0947	1	0.75823	
age	5.7134	1	0.01684	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
m2 <- glm(
  formula = survie ~ age + futime,
  data = donnees_survie,
  family = binomial
)
car::Anova(m2, type = "2")
```

Analysis of Deviance Table (Type II tests)

Response: survie

	LR	Chisq	Df	Pr(>Chisq)
age	5.6245	1	0.01771	*
futime	26.8655	1	2.181e-07	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

AIC(m1, m2)

	df	AIC
m1	4	94.82011
m2	3	92.91486

summary(m2)

Call:

glm(formula = survie ~ age + futime, family = binomial, data = donnees_survie)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
--	----------	------------	---------	----------

```

(Intercept) -0.5665844  1.1356419  -0.499   0.6178
age          0.0603137  0.0258221   2.336   0.0195 *
fuptime      -0.0030262  0.0007125  -4.248  2.16e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 120.528  on 102  degrees of freedom
Residual deviance:  86.915  on 100  degrees of freedom
AIC: 92.915

Number of Fisher Scoring iterations: 5

```

loi quasibinomiale en cas de surdispersion

```

m3 <- glm(
  formula = survie ~ age + fuptime,
  data = donnees_survie,
  family = quasibinomial
)

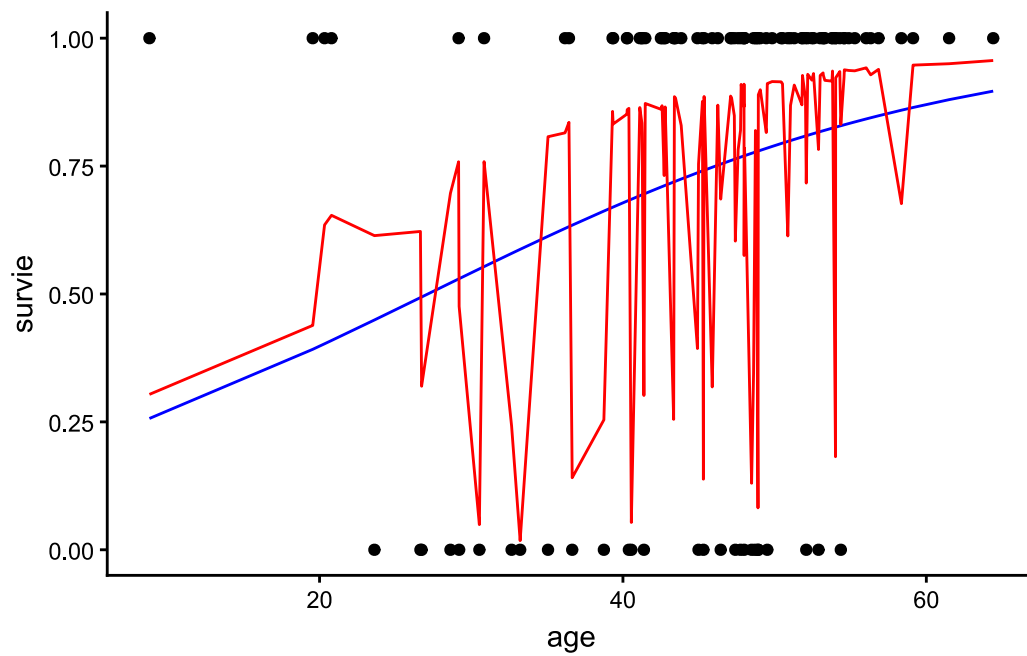
```

représentation graphique

```

donnees_survie |>
  mutate(
    prediction_survie_age_seul =
      glm(
        formula = survie ~ age,
        data = donnees_survie,
        family = binomial
      )$fitted.values,
    prediction_survie = m2$fitted.values
  ) |>
  ggplot() +
  aes(x = age, y = fustat) +
  geom_point() +
  geom_line(aes(y = prediction_survie_age_seul), color = "blue") +
  geom_line(aes(y = prediction_survie), color = "red") +
  ylab("survie") +
  theme_classic()

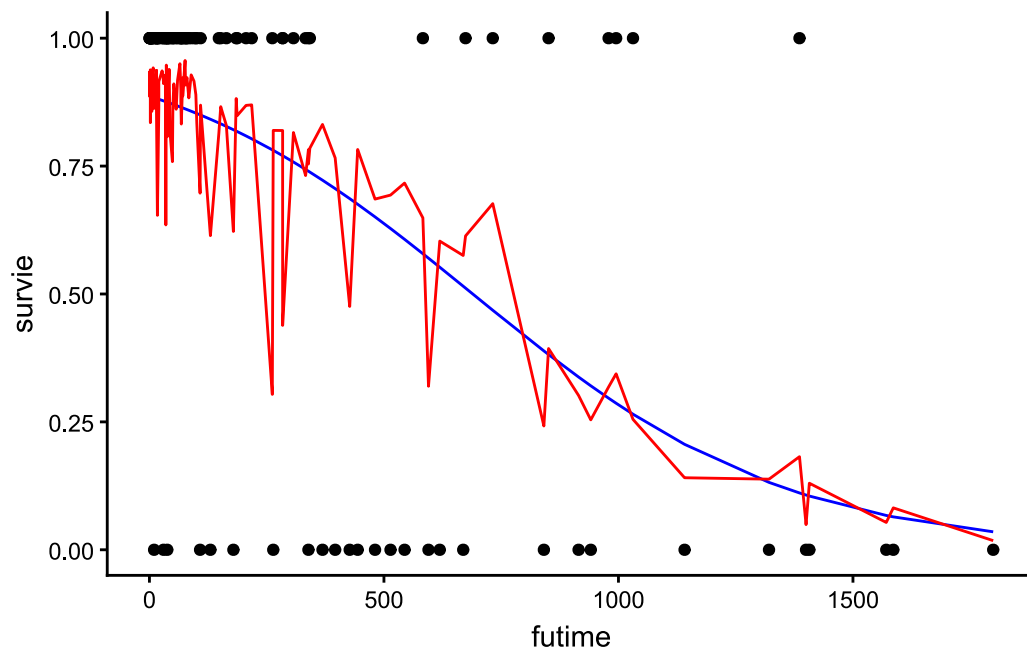
```



```

donnees_survie |>
  mutate(
    prediction_survie_futime_seul =
      glm(
        formula = survie ~ futime,
        data = donnees_survie,
        family = binomial
      )$fitted.values,
    prediction_survie = m2$fitted.values
  ) |>
  ggplot() +
  aes(x = futime, y = fustat) +
  geom_point() +
  geom_line(aes(y = prediction_survie_futime_seul), color = "blue") +
  geom_line(aes(y = prediction_survie), color = "red") +
  ylab("survie") +
  theme_classic()

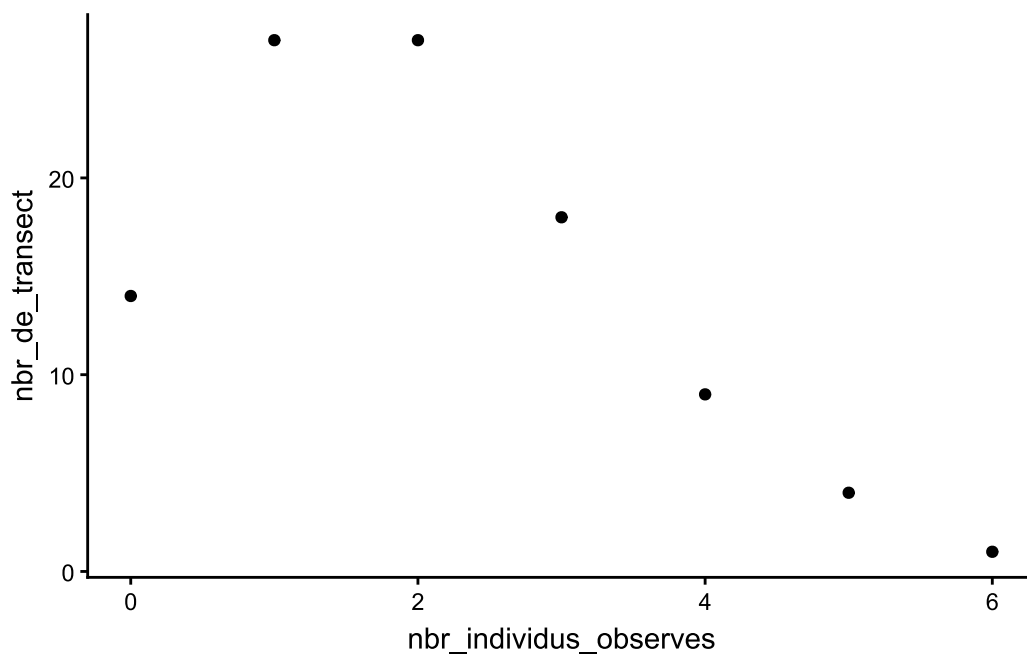
```



3.2 régression de poisson

3.2.a Exemple de la loi de Poisson

```
tibble(
  nbr_individus_observes = 0:6,
  nbr_de_transect = round(dpois(nbr_individus_observes, lambda = 2)*100)
) |>
ggplot() +
  aes(x = nbr_individus_observes, y = nbr_de_transect) +
  geom_point() +
  theme_classic()
```



3.2.b Loi de Poisson, données **Insurance**

```
declaration_accident <- MASS::Insurance
```

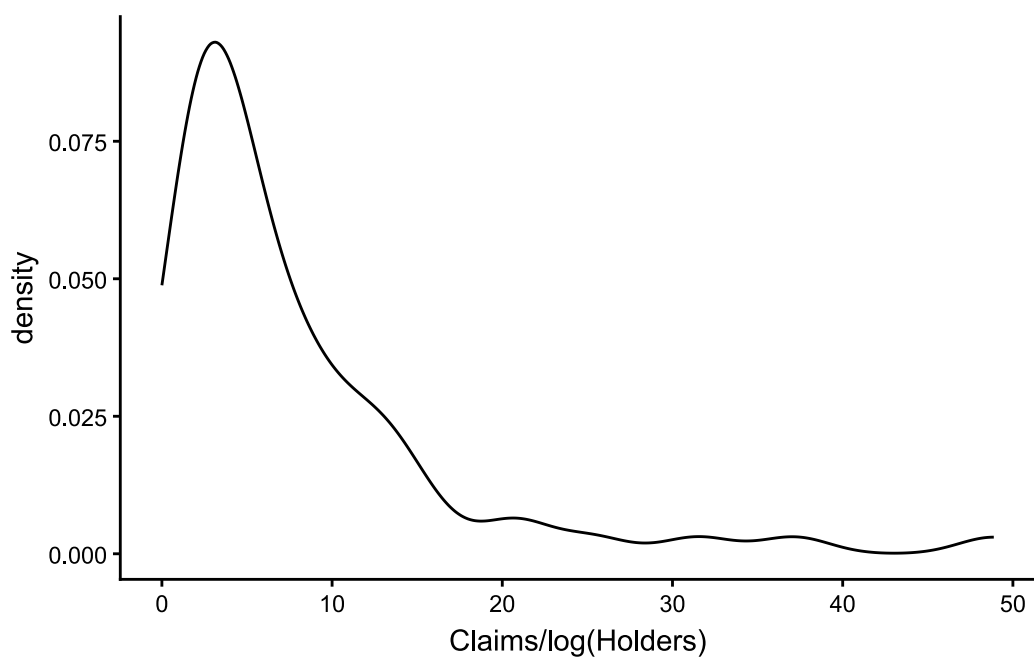
```
glimpse(declaration_accident)
```

```
Rows: 64
```

```
Columns: 5
```

```
$ District <fct> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2...
$ Group    <ord> <1l, <1l, <1l, <1l, 1-1.5l, 1-1.5l, 1-1.5l, 1-1.5l, 1.5-2l, 1...
$ Age      <ord> <25, 25-29, 30-35, >35, <25, 25-29, 30-35, >35, <25, 25-29, 3...
$ Holders  <int> 197, 264, 246, 1680, 284, 536, 696, 3582, 133, 286, 355, 1640...
$ Claims   <int> 38, 35, 20, 156, 63, 84, 89, 400, 19, 52, 74, 233, 4, 18, 19,...
```

```
ggplot(declaration_accident) +
  aes(x = Claims/log(Holders)) +
  geom_density() +
  theme_classic()
```



```
ggsave("img/08Rl13N.png", width = 14, height = 10, units = "cm")
```

```
m1 <- glm(
  Claims ~ Age*District*Group + offset(log(Holders)),
  data = declaration_accident,
  family = poisson)
car::Anova(m1)
```

Analysis of Deviance Table (Type II tests)

Response: Claims

	LR	Chisq	Df	Pr(>Chisq)
Age	84.736	3	< 2.2e-16	***
District	13.942	3	0.002985	**
Group	88.757	3	< 2.2e-16	***
Age:District	6.237	9	0.715940	
Age:Group	10.395	9	0.319432	
District:Group	7.167	9	0.619753	
Age:District:Group	27.290	27	0.448253	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
m2 <- glm(
  Claims ~ Age*District + Age*Group + Group*District + offset(log(Holders)),
  data = declaration_accident,
  family = poisson)
car::Anova(m2)
```

Analysis of Deviance Table (Type II tests)

Response: Claims

	LR	Chisq	Df	Pr(>Chisq)							
Age	84.736	3	< 2.2e-16	***							
District	13.942	3	0.002985	**							
Group	88.757	3	< 2.2e-16	***							
Age:District	6.237	9	0.715940								
Age:Group	10.395	9	0.319432								
District:Group	7.167	9	0.619753								

Signif. codes:	0	'***'	0.001	'**'	0.01	'*'	0.05	'.'	0.1	' '	1

```
m3 <- glm(
  Claims ~ Age*Group + Group*District + offset(log(Holders)),
  data = declaration_accident,
  family = poisson)
car::Anova(m3)
```

Analysis of Deviance Table (Type II tests)

Response: Claims

	LR	Chisq	Df	Pr(>Chisq)							
Age	84.736	3	< 2.2e-16	***							
Group	88.667	3	< 2.2e-16	***							
District	13.942	3	0.002985	**							
Age:Group	10.604	9	0.303797								
Group:District	7.380	9	0.597588								

Signif. codes:	0	'***'	0.001	'**'	0.01	'*'	0.05	'.'	0.1	' '	1

```
m4 <- glm(
  Claims ~ Age*Group + District + offset(log(Holders)),
  data = declaration_accident,
```

```
family = poisson)
car::Anova(m4)
```

Analysis of Deviance Table (Type II tests)

Response: Claims

	LR	Chisq	Df	Pr(>Chisq)
Age	84.870	3	< 2.2e-16	***
Group	88.667	3	< 2.2e-16	***
District	13.942	3	0.002985	**
Age:Group	10.513	9	0.310599	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
m5 <- glm(
  Claims ~ Age + Group + District + offset(log(Holders)),
  data = declaration_accident,
  family = poisson)
car::Anova(m5)
```

Analysis of Deviance Table (Type II tests)

Response: Claims

	LR	Chisq	Df	Pr(>Chisq)
Age	84.870	3	< 2.2e-16	***
Group	88.667	3	< 2.2e-16	***
District	13.871	3	0.003086	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
AIC(m1, m5)
```

	df	AIC
m1	64	445.3215
m5	10	388.7416

vérification des conditions d'utilisation

```
summary(m5)
```



```

Call:
glm(formula = Claims ~ Age + Group + District + offset(log(Holders)),
    family = poisson, data = declaration_accident)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.810508    0.032972 -54.910  < 2e-16 ***
Age.L        -0.394432    0.049404  -7.984 1.42e-15 ***
Age.Q        -0.000355    0.048918  -0.007 0.994210
Age.C        -0.016737    0.048478  -0.345 0.729910
Group.L       0.429708    0.049459   8.688  < 2e-16 ***
Group.Q       0.004632    0.041988   0.110 0.912150
Group.C      -0.029294    0.033069  -0.886 0.375696
District2     0.025868    0.043016   0.601 0.547597
District3     0.038524    0.050512   0.763 0.445657
District4     0.234205    0.061673   3.798 0.000146 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 236.26  on 63  degrees of freedom
Residual deviance:  51.42  on 54  degrees of freedom
AIC: 388.74

Number of Fisher Scoring iterations: 4

```

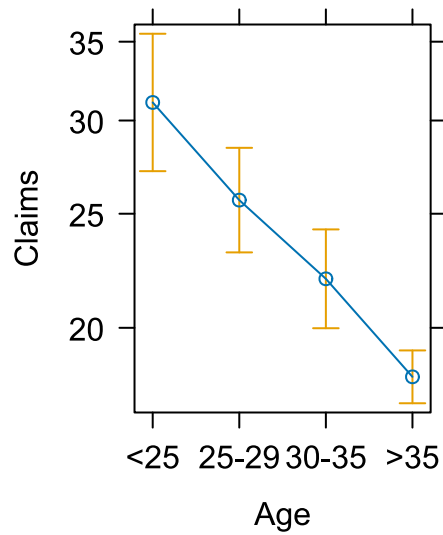
representaiton graphique des effets

```

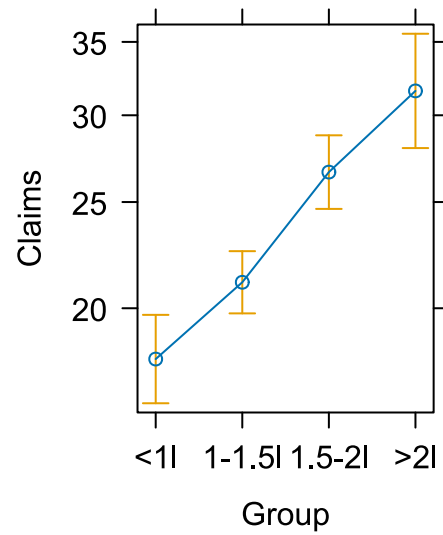
plot(
  effects::allEffects(m5)
)

```

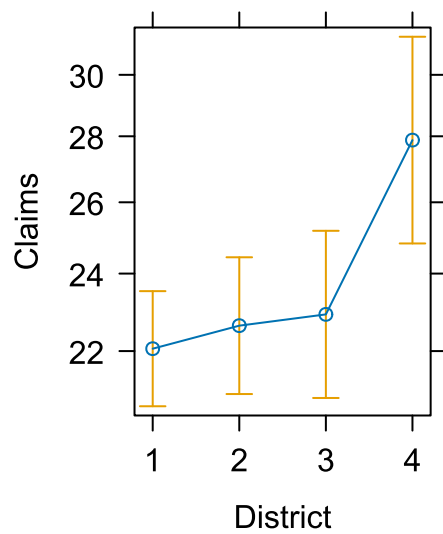
Age effect plot



Group effect plot

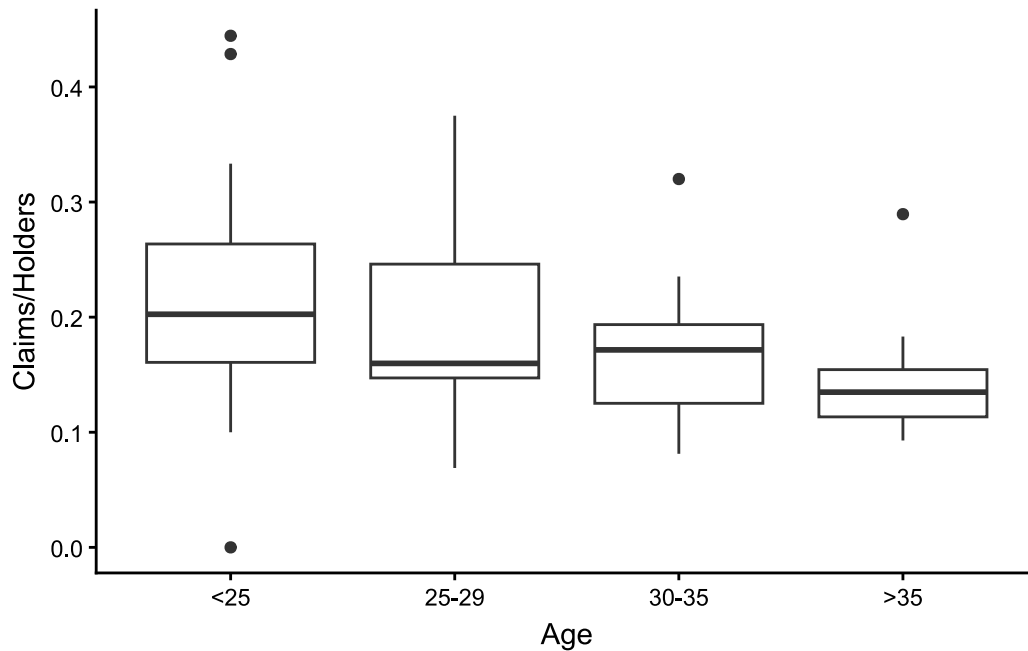


District effect plot



```
ggplot(declaration_accident) +  
  aes(  
    x = Age,  
    y = Claims/holders  
  ) +
```

```
geom_boxplot() +  
theme_classic()
```



4 modèle linéaire mixte

4.1 Prise de masse de rats sous trois régimes

```
library(nlme)
```

Attachement du package : 'nlme'

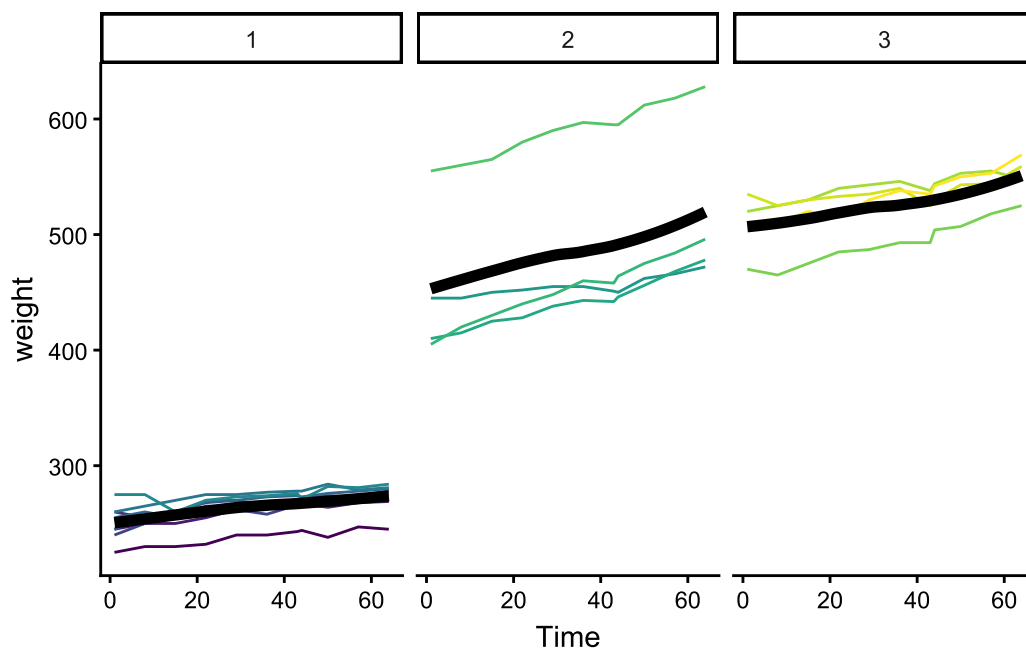
L'objet suivant est masqué depuis 'package:dplyr':

```
collapse
```

```
BodyWeight |>  
  ggplot() +  
  aes(x = Time, y = weight, color = Rat) +  
  geom_line() +  
  geom_smooth(se = FALSE, color = "black", linewidth = 2) +  
  facet_wrap(vars(Diet)) +
```

```
theme_classic() +
theme(legend.position = "none")
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



```
ggsave("img/08Rl18N.png", width = 14, height = 14, units = "cm")
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
glimpse(BodyWeight)
```

```
Rows: 176
Columns: 4
$ weight <dbl> 240, 250, 255, 260, 262, 258, 266, 266, 265, 272, 278, 225, 230...
$ Time   <dbl> 1, 8, 15, 22, 29, 36, 43, 44, 50, 57, 64, 1, 8, 15, 22, 29, 36,...
$ Rat    <ord> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, ...
$ Diet   <fct> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
```

4.1.a réalisation du modèle linéaire mixte avec {nlme}

```
lme1 <- nlme::lme(weight ~ Diet*Time, random = ~ 1|Rat, data = BodyWeight)
car::Anova(lme1, type = "3")
```

Analysis of Deviance Table (Type III tests)

Response: weight

	Chisq	Df	Pr(>Chisq)
(Intercept)	373.88	1	< 2.2e-16 ***
Diet	155.17	2	< 2.2e-16 ***
Time	105.02	1	< 2.2e-16 ***
Diet:Time	102.17	2	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
summary(lme1)
```

Linear mixed-effects model fit by REML

Data: BodyWeight

	AIC	BIC	logLik
	1248.245	1273.332	-616.1226

Random effects:

Formula: ~1 | Rat

(Intercept) Residual

StdDev: 36.60911 6.404333

Fixed effects: weight ~ Diet * Time

	Value	Std.Error	DF	t-value	p-value
(Intercept)	251.65165	13.014623	157	19.336069	0
Diet2	200.66549	22.541989	13	8.901854	0
Diet3	252.07168	22.541989	13	11.182318	0
Time	0.35964	0.035095	157	10.247714	0
Diet2:Time	0.60584	0.060786	157	9.966824	0
Diet3:Time	0.29834	0.060786	157	4.908031	0

Correlation:

	(Intr)	Diet2	Diet3	Time	Dt2:Tm
Diet2	-0.577				
Diet3	-0.577	0.333			
Time	-0.090	0.052	0.052		
Diet2:Time	0.052	-0.090	-0.030	-0.577	

```
Diet3:Time  0.052 -0.030 -0.090 -0.577  0.333

Standardized Within-Group Residuals:
      Min       Q1       Med       Q3       Max
-2.7269865 -0.4678802  0.0070912  0.4762223  3.3603548

Number of Observations: 176
Number of Groups: 16
```

test post-hoc {multcomp}

```
summary(
  glht(
    lme1,
    linfct = mcp(
      Diet = "Tukey"
    )
  )
)
```

Warning in mcp2matrix(model, linfct = linfct): covariate interactions found -- default contrast might be inappropriate

```
Simultaneous Tests for General Linear Hypotheses

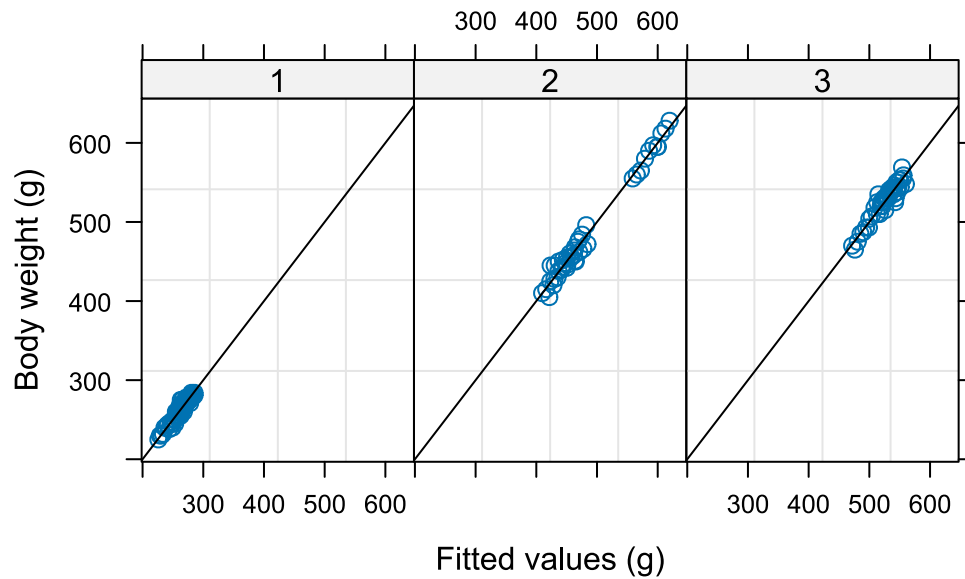
Multiple Comparisons of Means: Tukey Contrasts


Fit: lme.formula(fixed = weight ~ Diet * Time, data = BodyWeight,
  random = ~1 | Rat)

Linear Hypotheses:
              Estimate Std. Error z value Pr(>|z|)
2 - 1 == 0    200.67      22.54   8.902  <0.001 ***
3 - 1 == 0    252.07      22.54  11.182  <0.001 ***
3 - 2 == 0     51.41      26.03   1.975   0.118
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)
```

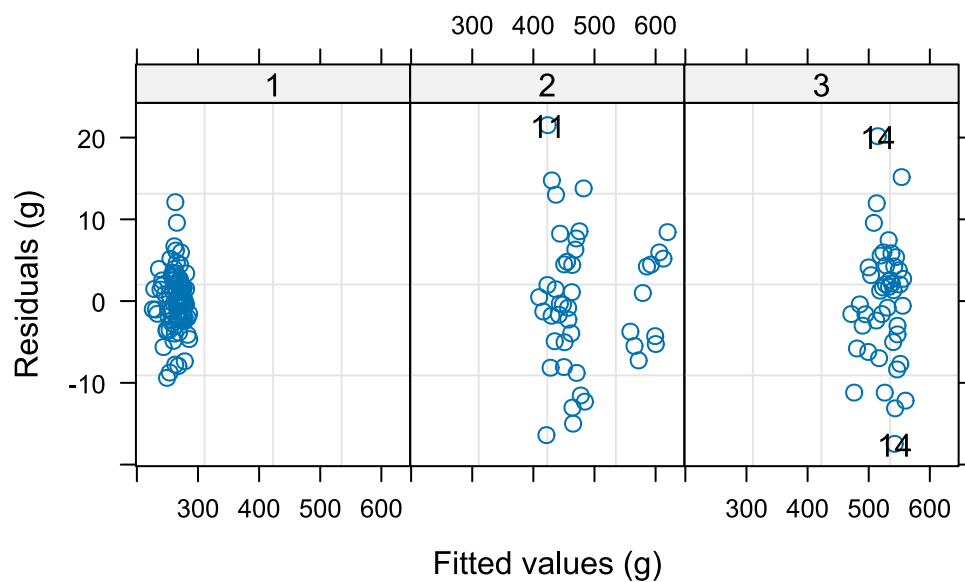
vérification de la validité du modèle Représentativité globale

```
nlme::plot.lme(lme1, weight ~ fitted(.)|Diet, abline = c(0,1))
```



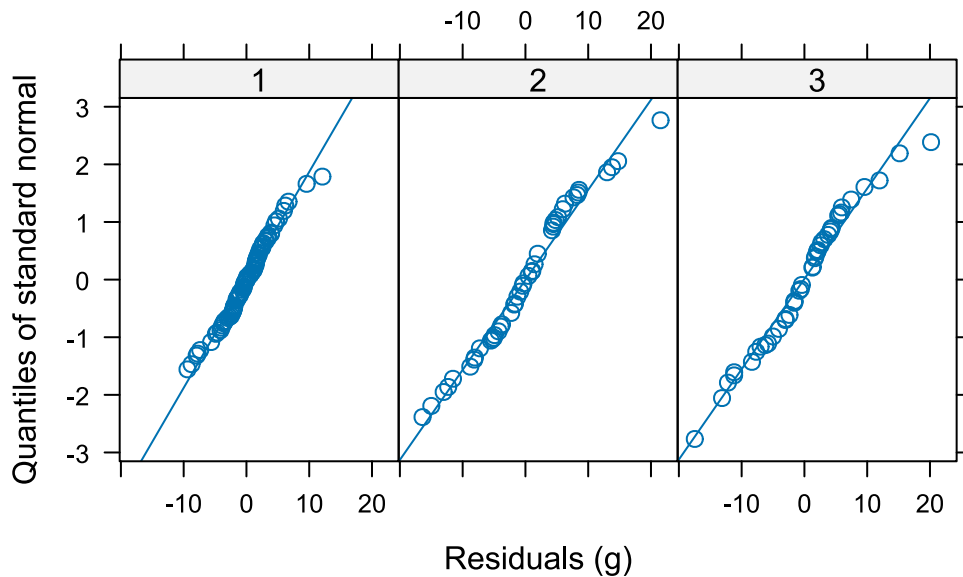
homogénéité des résidus

```
nlme::plot.lme(lme1, resid(.) ~ fitted(.)|Diet, id = 0.01)
```



normalité des résidus

```
qqnorm(lme1, ~ resid(.)|Diet, type = c("p", "r"))
```



4.1.b réalisation du modèle linéaire mixte avec {lme4}

```
lmer1 <- lme4::lmer(weight ~ Diet * Time + (1|Rat) , data = BodyWeight)  
car::Anova(lmer1, type = "3")
```

Analysis of Deviance Table (Type III Wald chisquare tests)

Response: weight

	Chisq	Df	Pr(>Chisq)
(Intercept)	373.88	1	< 2.2e-16 ***
Diet	155.17	2	< 2.2e-16 ***
Time	105.02	1	< 2.2e-16 ***
Diet:Time	102.17	2	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
summary(lmer1)
```



```
Linear mixed model fit by REML ['lmerMod']
Formula: weight ~ Diet * Time + (1 | Rat)
Data: BodyWeight
```

REML criterion at convergence: 1232.2

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.7270	-0.4679	0.0071	0.4762	3.3604

Random effects:

Groups	Name	Variance	Std.Dev.
Rat	(Intercept)	1340.23	36.609
	Residual	41.02	6.404

Number of obs: 176, groups: Rat, 16

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	251.65165	13.01462	19.336
Diet2	200.66549	22.54199	8.902
Diet3	252.07168	22.54199	11.182
Time	0.35964	0.03509	10.248
Diet2:Time	0.60584	0.06079	9.967
Diet3:Time	0.29834	0.06079	4.908

Correlation of Fixed Effects:

	(Intr)	Diet2	Diet3	Time	Dt2:Tm
Diet2	-0.577				
Diet3	-0.577	0.333			
Time	-0.090	0.052	0.052		
Diet2:Time	0.052	-0.090	-0.030	-0.577	
Diet3:Time	0.052	-0.030	-0.090	-0.577	0.333

test post-hoc

```
summary(
  glht(
    lmer1,
    linfct = mcp(
      Diet = "Tukey"
    )
  )
)
```

```
Warning in mcp2matrix(model, linfct = linfct): covariate interactions found --
default contrast might be inappropriate
```

Simultaneous Tests for General Linear Hypotheses

Multiple Comparisons of Means: Tukey Contrasts

```
Fit: lme4::lmer(formula = weight ~ Diet * Time + (1 | Rat), data = BodyWeight)
```

Linear Hypotheses:

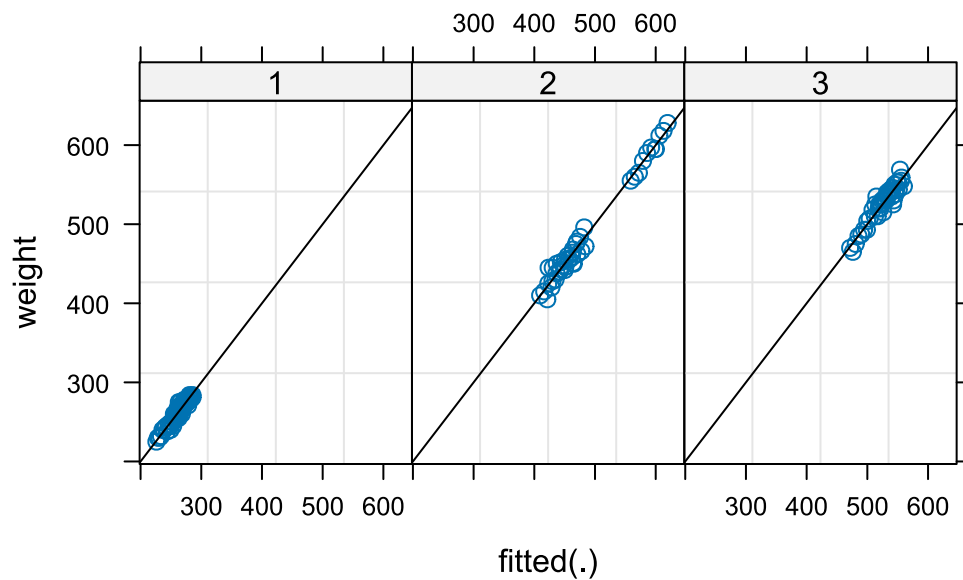
	Estimate	Std. Error	z value	Pr(> z)
2 - 1 == 0	200.67	22.54	8.902	<0.001 ***
3 - 1 == 0	252.07	22.54	11.182	<0.001 ***
3 - 2 == 0	51.41	26.03	1.975	0.118

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Adjusted p values reported -- single-step method)

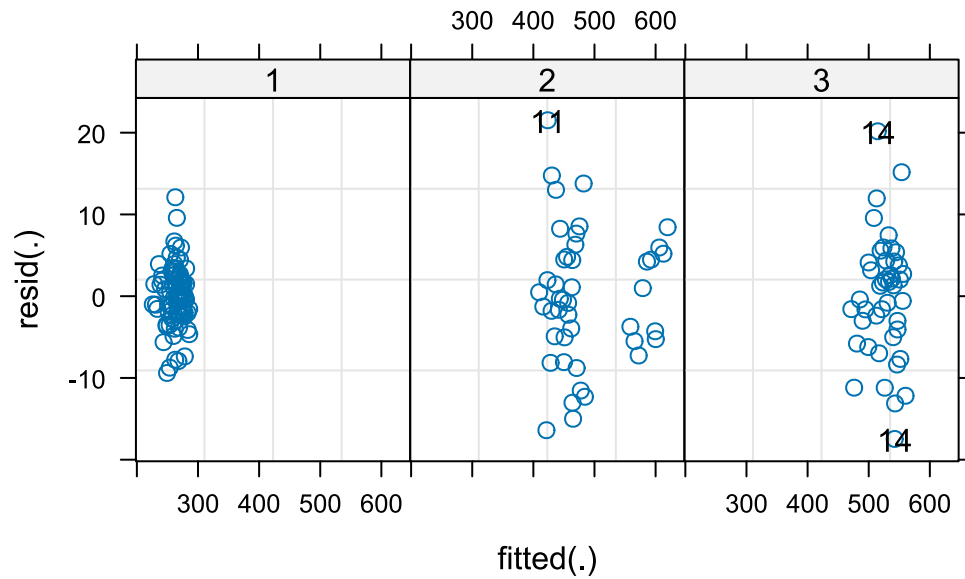
vérification de la validité du modèle Représentativité globale

```
plot(lmer1, weight ~ fitted(.)|Diet, abline = c(0,1))
```



homogénéité des résidus

```
plot(lmer1, resid(.) ~ fitted(.)|Diet, id = 0.01)
```



normalité des résidus

```
tibble(  
  residus = residuals(lmer1),  
  regime = BodyWeight$Diet  
) |>  
  ggplot() +  
  aes(sample = residus) +  
  geom_qq() +  
  geom_qq_line(color = "dark red", linewidth = 1) +  
  facet_wrap(~ regime, ncol = 3) +  
  theme_classic()
```

